

Bio Informatics

Lecture 15

Multiple Sequence Alignments – Progressive Alignments

- This method, also known as the **hierarchical** or **tree method**, was developed by **Paulien Hogeweg** and **Ben Hesper** in 1984. It builds up a final **MSA** by combining pairwise alignments beginning with the most similar pair and progressing to the most distantly related pair.
- Two of the popular progressive alignment methods used at present are,
- [Clustal Omega](#)
- [T-Coffee](#)

Progressive Alignments - Example

- Seq_1: ACCGTGAAGCCAATAC
- Seq_2: ACGTGCAACCATTAC
- Seq_3: AGCGTGCAGCCAATAC
- Seq_4: AGGGTGCCGCAATAC
- Seq_5: AGGGTGCCACAATAC